

Figure 1 is a plot showing the relative error of the second virial coefficient, B_2 , as a function of the relative error of the second virial coefficient, B_2 . The x-axis is labeled "Relative error of B_2 " and ranges from 0 to 1. The y-axis is labeled "Relative error of B_2 " and ranges from 0 to 1. The plot displays several data series, each representing a different computational method:

- HF-D4 (I) (Blue line with circles)
- SAPT(PBE0)-D4 (S) (Orange line with circles)
- SAPT(PBE0)-D4 (I) (Green line with circles)
- SAPT(PBE0)-D4 (S, ND) (Red line with circles)
- SAPT(PBE0)-D4 (I, ND) (Purple line with circles)
- SAPT(PBE0) (Grey line with circles)
- SAPT2+3(CCD)/aTZ (Black line with circles)

The plot shows that the relative error of B_2 for the HF-D4 (I) method is significantly higher than for the other methods, which are clustered together at lower relative errors. The SAPT2+3(CCD)/aTZ method shows the lowest relative error, followed by SAPT(PBE0)-D4 (S, ND), SAPT(PBE0)-D4 (I, ND), SAPT(PBE0)-D4 (S), SAPT(PBE0)-D4 (I), SAPT(PBE0), and finally HF-D4 (I).